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Student Number

2015

HALF YEARLY
EXAMINATION

Extension 1 Mathematics

20th March 2015

General Instructions

- Reading time – 5 minutes
- Working time 2 hours
- Write using blue or black pen
Black pen is preferred
- Approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11-14 show relevant mathematical reasoning and/or calculations
- Start a new booklet for each question

Total Marks – 70

Section I - Pages 2 - 6

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II - Pages 7 - 10

60 marks

- Attempt Questions 11 – 14
- Allow about 1 hour and 45 minutes for this section

Question	Marks
1 - 10	/10
11	/15
12	/15
13	/15
14	/15
Total	/70

THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM

This assessment task constitutes 25% of the Higher School Certificate Course Assessment

Section I

10 marks

Attempt Questions 1 – 10

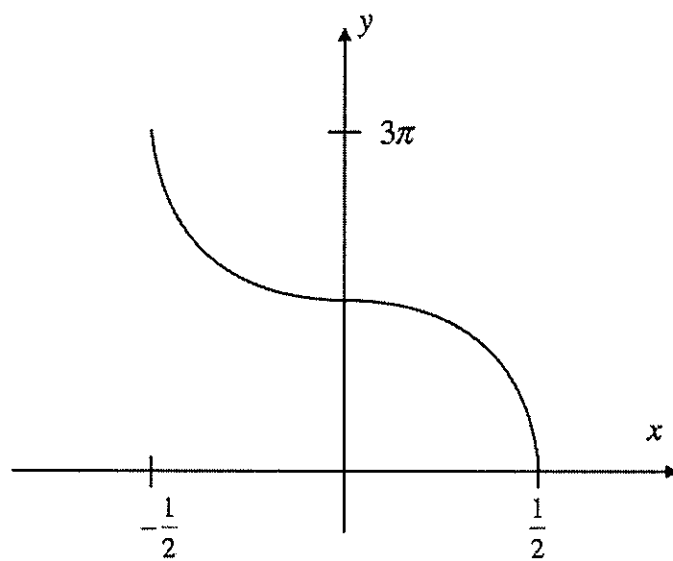
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for questions 1 – 10 (Detach from paper)

- 1 The value of k , if $x+1$ is a factor of $x^3 - 2x^2 - kx + 5$ is:
- (A) 4
 - (B) 2
 - (C) -2
 - (D) -6

- 2 The solution to $x(x-2)^2(3+x)^3 < 0$ is:
- (A) $-3 < x < 0$
 - (B) $x < -3$ or $x > 0$
 - (C) $x < -3$ or $x > 2$
 - (D) $-3 < x < 2$

3



The graph sketched above represents:

(A) $y = 3 \cos^{-1} 2x$

(B) $y = 3 \sin^{-1} \frac{x}{2}$

(C) $y = \cos^{-1} \frac{x}{3}$

(D) $y = 3 \cos^{-1} \frac{x}{2}$

4 What is the exact value of $\sin\left(2\cos^{-1}\frac{3}{4}\right)$

(A) $\frac{3\sqrt{7}}{8}$

(B) $\frac{15}{8}$

(C) $\frac{\sqrt{7}}{2}$

(D) $\frac{21}{8}$

5 What is the gradient of the tangent to the curve $y = \sin^{-1} 2x$ at the point where $x = -\frac{1}{4}$

(A) $\frac{\sqrt{56}}{7}$

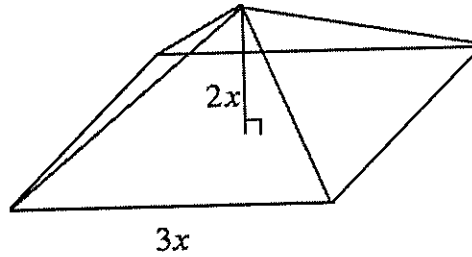
(B) $\frac{4\sqrt{3}}{3}$

(C) $\frac{2\sqrt{3}}{3}$

(D) $\frac{8\sqrt{3}}{3}$

- 6 Two roots of the equation $2x^3 + x^2 - px + 6 = 0$ are reciprocals of one another. The value of p is:
- (A) 13
(B) 23
(C) -13
(D) -23
- 7 What is the exact value of $\sin 36^\circ \cos 24^\circ + \cos 36^\circ \sin 24^\circ$?
- (A) $\frac{1}{\sqrt{2}}$
(B) $\frac{\sqrt{3}}{2}$
(C) $\frac{1}{2}$
(D) $\sqrt{3}$
- 8 What can the expression $\frac{1 + \cos x + \cos 2x}{\sin x + \sin 2x}$ be simplified to:
- (A) $1 + \cot x + \cot 2x$
(B) $\cot x - \operatorname{cosec} x$
(C) $\cot x$
(D) $\operatorname{cosec} 3x + \cot 3x$

- 9 The diagram shows a square based pyramid. The ratio of the perpendicular height to the side of the base is 2:3. The angle between a side face and the base is closest to:



- (A) 38°
 (B) 43°
 (C) 53°
 (D) 59°
- 10 What is the exact value of $\int_0^{\frac{\pi}{8}} \sin^2 2x dx$

- (A) $\frac{\pi}{16}$
 (B) $\frac{1}{16}(\pi - 2)$
 (C) $\frac{1}{16}(\pi - 8)$
 (D) $\frac{1}{16}(\pi + 8)$

Section II

60 marks

Attempt Questions 11 – 14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11(15 marks) Use a SEPARATE writing booklet

- (a) Solve $\frac{2x+1}{x-1} \geq 3, \quad x \neq 1$ 2
- (b) Differentiate
- (i) e^{6x+3} 1
- (ii) $e^{2x^2} + \log 3x$ 2
- (c) Simplify $\int \frac{6x+12}{x^2+4x} dx$ 2
- (d) Simplify $\int \sin x \cos^2 x dx$ 2
- (e) Simplify $\int 5^x dx$ 2
- (f) Evaluate $\int_0^3 \frac{1}{9+x^2} dx$ 2
- (g) If α, β, γ and δ are the roots of the equation $x^4 - 3x^3 + x - 7 = 0$, find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}$ 2

End of Question 11

Question 12(15 marks) Use a SEPARATE writing booklet

- (a) Using the substitution $u = 1+t$ find: 3

$$\int \frac{t}{\sqrt{1+t}} dt$$

- (b) Using the substitution $u = 2-x$ evaluate: 3

$$\int_1^2 x\sqrt{2-x} dx$$

- (c) Show that $\frac{x+7}{x-1}$ can be written as $1 + \frac{8}{x-1}$ and hence show that 3

$$\int_2^{e+1} \frac{x+7}{x-1} dx = e+7$$

- (d) Using the result $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, find the value of $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$ 3

- (e) The function $h(x)$ is given by

$$h(x) = \sin^{-1} x + \cos^{-1} x, \quad 0 \leq x \leq 1$$

- i.) Find $h'(x)$ 1

- ii.) Sketch the graph of $y = h(x)$ 2

End of Question 12

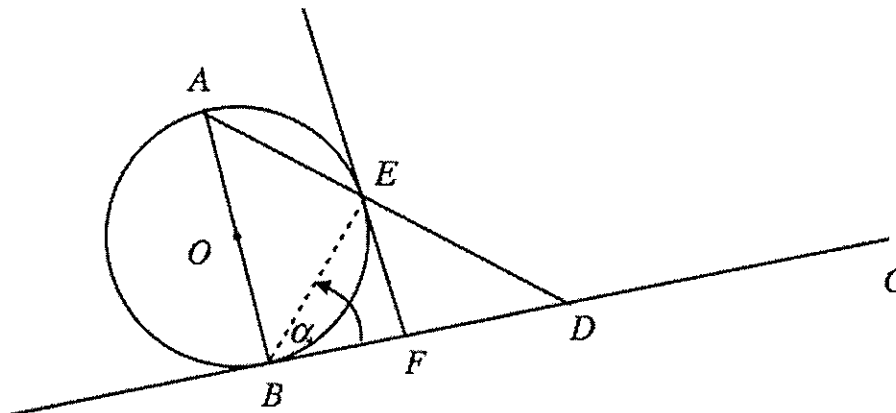
Question 13(15 marks) Use a SEPARATE writing booklet

- (a) Find the volume of the solid generated when the area bounded by the curve $y = 1 + \sin x$, the x axis and the ordinates $x = 0, x = 2\pi$, is rotated about the x axis. 4

- (b) Evaluate $\int_0^{\sqrt{2}} \frac{\sqrt{2}}{\sqrt{4-2x^2}} dx$ 3

- (c) Express $\sqrt{3} \sin x - \cos x$ in the form $r \sin(x - \alpha)$ and hence or otherwise solve $\sqrt{3} \sin x - \cos x = 1$ for $0 \leq x \leq 2\pi$ 4

- (d)



In the diagram, AB is a diameter of the circle, centre O, and BC is a tangent to the circle at B. The line AED intersects the circle at E and BC at D. The tangent to the circle at E intersects BC at F. Let $\angle EBF = \alpha$ radians.

Copy the diagram into your writing booklet.

- (i) Prove $\angle FED = \frac{\pi}{2} - \alpha$ 2

- (ii) Prove that $BF = FD$ 2

End of Question 13

Question 14(15 marks) Use a SEPARATE writing booklet

- (a) (i) Sketch the graphs of $y = 2 \sin x$ and $y = x$ on the same set of axes. 2
Show that $x - 2 \sin x = 0$ has a root near $x = 2$.
- (ii) Using one application of Newton's method, find a better approximation, correct to 2 decimal places. 2
- (b) $P(2ap, ap^2)$ is a point on the parabola $x^2 = 4ay$.
- i.) Show that the equation of the normal to the parabola at the point P is: 2
$$x + py = 2ap + ap^3$$
- ii.) Find the coordinates of the point Q where the normal at P meets the y -axis. 1
- iii.) Show that the point R which divides PQ **externally** in the ratio 2: 1 2
has coordinates $(-2ap, 4a + ap^2)$
- iv.) Find the Cartesian equation of the locus of R . Show that the locus is 3
a parabola and write down the coordinates of its vertex and focus.
- (c) The straight lines $3x - 5y - 1 = 0$, $bx - 2y + 2 = 0$ are inclined to each other 3
at an angle of 45° . Find the possible values of b .

End of Question 14

End of Assessment

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$



Student Number

YR. 12. MATHEMATICS EXTENSION 1. (1)

M17 COURSE ASSESSMENT 2015.

SOLUTIONS / MARKING SCHEME

SECTION 1.

QUESTIONS 1-10

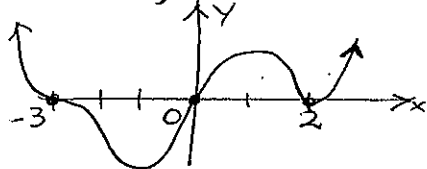
SUMMARY.

1. K 2. A 3. A 4. A 5. B
6. A 7. B 8. C 9. C 10. B

SOLUTIONS.

1. $(x+1)$ is a factor then $P(-1) = 0$
 $P(-1) = (-1)^3 - 2(-1)^2 - kx - 1 + 5$
 $\therefore 0 = -1 - 2 + k + 5$
 $-2 = k$ (C)

2. $x(x-2)^2(3+x)^3 < 0$

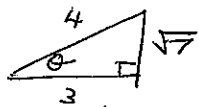


$\therefore -3 < x < 0$ (A)

3. $-1 < 2x < 1$; $0 \leq y \leq \pi$
 $-\frac{1}{2} < x < \frac{1}{2}$; $0 \leq 3y \leq 3\pi$
 $\cos^{-1} x$

$\therefore y = 3 \cos^{-1} 2x$ (A)

4. $\sin(2 \cos^{-1} \frac{3}{4})$
 $\sin 2\theta$
 $= 2 \sin \theta \cos \theta$
 $= 2 \times \frac{\sqrt{7}}{4} \times \frac{3}{4} = \frac{3\sqrt{7}}{8}$ (A)



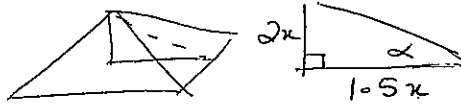
1 mark each
= 10 marks.

5. $y = \sin^{-1} 2x$
 $y' = \frac{1}{\sqrt{1-(2x)^2}} \cdot 2$
 $= \frac{2}{\sqrt{1-4x^2}}$
 $f'(-\frac{1}{4}) = \frac{2}{\sqrt{1-4 \times \frac{1}{16}}}$
 $= \frac{2}{\sqrt{3/4}}$
 $= 2 \times \frac{2}{\sqrt{3}}$
 $= \frac{4}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$
 $= \frac{4\sqrt{3}}{3}$ (B)

6. $\alpha, \beta, \frac{1}{\beta}$; $2x^3 + x^2 - px + 6 = 0$
 $\therefore$ product $= \alpha$
 $\therefore \frac{-d}{a} = \frac{-6}{2} = -3$
 $\therefore \alpha = -3$
 subst. $\alpha = -3$
 $2(-3)^3 + (-3)^2 - px - 3 + 6 = 0$
 $-54 + 9 + 3p + 6 = 0$
 $3p = 39$
 $p = 13$ (A)

7. $\sin 36^\circ \cos 24^\circ + \cos 36^\circ \sin 24^\circ$
 $= \sin(36+24)$
 $= \sin 60^\circ$
 $= \frac{\sqrt{3}}{2}$ (B)

8. $\frac{1 + \cos x + \cos 2x}{\sin x + \sin 2x}$
 $= \frac{1 + \cos x + 2\cos^2 x - 1}{\sin x + 2\sin x \cos x}$
 $= \frac{\cos x (2\cos x + 1)}{\sin x (2\cos x + 1)}$
 $= \cot x$ (C)

9. 
 $\therefore \tan \alpha = \frac{2x}{1.5x}$
 $\tan \alpha = \frac{4}{3}$
 $\alpha = 53^\circ$ (nearest $^\circ$) (C)

10. $\int_0^{\pi/8} \sin^2 2x \cdot dx$
 using $\cos 2\theta = 1 - 2\sin^2 \theta$
 $\cos 4\theta = 1 - 2\sin^2 2\theta$
 $2\sin^2 2\theta = 1 - \cos 4\theta$
 $\sin^2 2\theta = \frac{1}{2}(1 - \cos 4\theta)$
 expression becomes:
 $= \frac{1}{2} \int_0^{\pi/8} 1 - \cos 4x \cdot dx$
 $= \frac{1}{2} \left[x - \frac{\sin 4x}{4} \right]_0^{\pi/8}$
 $= \frac{1}{2} \left[\frac{\pi}{8} - \frac{\sin \pi/2}{4} - 0 - 0 \right]$
 $= \frac{1}{2} \left[\frac{\pi}{8} - \frac{1}{4} \right]$
 $= \frac{\pi}{16} - \frac{1}{8}$
 $= \frac{1}{16}(\pi - 2)$ (B)

QUESTION 11

a) $\frac{2x+1}{x-1} \geq 3, x \neq 1$

$$x(x-1)^2$$

$$(x-1)(2x+1) \geq 3(x-1)^2$$

$$2x^2 - x - 1 \geq 3x^2 - 6x + 3$$

$$0 \geq x^2 - 5x + 4$$

$$(x-4)(x-1) \leq 0$$



$$1 \leq x \leq 4 \text{ but } x \neq 1 \therefore 1 < x \leq 4$$

This question could be done using 2 cases - award 1 mark for 1 case correct and then award 1 mark for correct answer

1 for correct inequality

some had difficulty in solving the quadratic inequality

1 < x, x ≤ 4 not accepted
1 for correct answer

b) i) $\frac{d}{dx} e^{6x+3} = 6e^{6x+3}$

1 mark correct answer

some did not differentiate log 3x properly

ii) $\frac{d}{dx} e^{2x^2} + \log 3x = 4xe^{2x^2} + \frac{1}{3x}$

$$= 4xe^{2x^2} + \frac{1}{3x}$$

1 mark for $4xe^{2x^2}$
1 mark for $\frac{1}{3x}$

c) $\int \frac{6x+12}{x^2+4x} \cdot dx = \int \frac{3(2x+4)}{x^2+4x} \cdot dx$

$$= 3 \ln(x^2+4x) + c$$

↑
some put $\frac{1}{3}$ instead of 3

1 mark for $\ln(x^2+4x)$

1 mark awarded for $3 \ln(x^2+4x)$ (no penalty for omitting 'c')

d) $\int \sin x \cdot \cos^2 x \cdot dx$

think: $\frac{d}{dx} \cos^3 x = 3 \cos^2 x \cdot -\sin x$

$$\therefore \int \sin x \cdot \cos^2 x \cdot dx = -\frac{1}{3} \cos^3 x + c$$

1 mark for $\cos^3 x$

1 mark for correct answer

QUESTION 11 continued.

e) $\int 5^x \cdot dx$

using $\frac{d}{dx} 5^x = \ln 5 \cdot 5^x$

$$\therefore \int 5^x \cdot dx = \frac{1}{\ln 5} \cdot 5^x + c$$

many did not give correct answer

1 mark for 5^x
1 mark for correct answer (no penalty for no 'c')

f) $\int_0^3 \frac{1}{9-x^2} dx = \left[\tan^{-1} \frac{x}{3} \right]_0^3$ wrong

deletion

$$= \tan^{-1} \frac{3}{3} - \tan^{-1} \frac{0}{3}$$

$$= \pi/4 + c$$

1 mark for correct integral
1 mark for $\pi/4$ (no penalty for no 'c')

g) $x^4 - 3x^3 + x - 7 = 0$

$$\frac{1}{2} + \frac{1}{3} + \frac{1}{8} + \frac{1}{8}$$

$$= \frac{384 + 256 + 128 + 128}{2304}$$

$$= -\frac{d}{a}$$

$$= \frac{-1}{a}$$

$$= \frac{-1}{-7}$$

$$= \frac{1}{7}$$

1 mark for correct algebraic statement

1 mark for correct answer

QUESTION 12

a) $u = 1+t$

$$\int \frac{t}{\sqrt{1+t}} dt. \quad u = 1+t$$

$$\frac{du}{dt} = 1$$

$$du = dt$$

Expression becomes

$$\int \frac{u-1}{\sqrt{u}} du$$

$$= \int \frac{u}{\sqrt{u}} - \frac{1}{\sqrt{u}} du$$

$$= \int u^{\frac{1}{2}} - u^{-\frac{1}{2}} du$$

$$= \frac{2}{3} u^{\frac{3}{2}} - 2u^{\frac{1}{2}} + C$$

$$= \frac{2}{3} (1+t)^{\frac{3}{2}} - 2\sqrt{1+t} + C$$

many students left their solution in terms of u

1 for correct substituted expression.

1 for correct integration
1 for correct expression in (t)

b) $\int_1^2 x\sqrt{2-x} dx$ $u = 2-x$, ~~$x=1$~~
also $x=1, u=1$ $\frac{du}{dx} = -1$
 $x=2, u=0$ $dx = -du$

Expression becomes

$$= -\int_1^0 (2-u)\sqrt{u} du$$

$$= -\int_0^1 (2-u)\sqrt{u} du$$

$$= -\left[\frac{4}{3} u^{\frac{3}{2}} - \frac{2}{5} u^{\frac{5}{2}} \right]_0^1$$

$$= -\left(0 - \left(\frac{4}{3} - \frac{2}{5} \right) \right)$$

$$= -\frac{-14}{15}$$

$$= \frac{14}{15}$$

many got the wrong way round the bounds

1 for correct expression in terms of u

1 for correct integration

1 for correct answer value.

5

12 c)

$$\frac{x+7}{x-1} = \frac{x-1+8}{x-1}$$

$$= \frac{x-1}{x-1} + \frac{8}{x-1}$$

$$= 1 + \frac{8}{x-1}$$

1 for showing

$$\int_2^{e+1} \frac{x+7}{x-1} dx = \int_2^{e+1} 1 dx + \int_2^{e+1} \frac{8}{x-1} dx$$

$$= [x]_2^{e+1} + [8 \ln(x-1)]_2^{e+1}$$

$$= (e+1-2) + (8 \ln(e) - 8 \ln(1))$$

$$= e-1+8$$

$$= e+7$$

$$= 14+5$$

1 mark for correct primitive

1 mark for correct substitution

12d) $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2} = \lim_{x \rightarrow 0} \frac{2 \sin^2 x}{x^2}$ Since $\cos 2x = 1 - 2 \sin^2 x$
 $\therefore 2 \sin^2 x = 1 - \cos 2x$

$$= 2 \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \frac{\sin x}{x}$$

$$= 2 \times 1$$

$$= 2$$

1 mark for trig substitution

1 mark for separating into standard limit

1 mark correct evaluation

12 e) $h(x) = \sin^{-1}(x) + \cos^{-1}(x)$

i) $\therefore h'(x) = \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x^2}}$

$$= 0$$

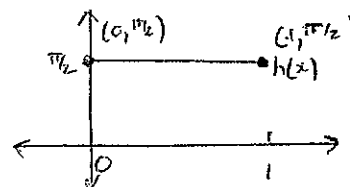
1 mark correct evaluation of $h'(x)$

ii) Since $h'(x) = 0 \rightarrow h(x)$ is a horizontal line

$$h(0) = \pi/2, \quad h(1) = \pi/2$$

1 mark working leading to conclusion that it is a horizontal line

1 mark correct graph with end points marked



$$y = 1 + \sin x$$

$$y^2 = 1 + 2\sin x + \sin^2 x$$

$$y^2 = \frac{3}{2} + 2\sin x + \frac{1}{2} - \frac{1}{2}\cos 2x$$

$$= \frac{3}{2} + 2\sin x - \frac{1}{2}\cos 2x$$

$$V = \pi \int_0^{2\pi} y^2 dx$$

$$= \pi \int_0^{2\pi} \left(\frac{3}{2} + 2\sin x - \frac{1}{2}\cos 2x \right) dx$$

$$= \pi \left[\frac{3}{2}x - 2\cos x + \frac{1}{4}\sin 2x \right]_0^{2\pi}$$

$$= \pi \left[\left(3\pi - 2 + 0 \right) - \left(0 - 2 + 0 \right) \right]$$

$$= \pi (3\pi)$$

$$= 3\pi^2$$

$$2\sin^2 x = 1 - \cos 2x$$

$$\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos 2x$$

QUESTION 13.

a) $y = 1 + \sin x \quad 0 \leq x \leq 2\pi$

$$V = \pi \int_0^{2\pi} y^2 dx$$



$$= \pi \int_0^{2\pi} (1 + \sin x)^2 dx$$

$$= \pi \int_0^{2\pi} (1 + 2\sin x + \sin^2 x) dx$$

$$= \pi \int_0^{2\pi} \left(1 + 2\sin x + \frac{1}{2}(1 - \cos 2x) \right) dx$$

$$= \pi \left[x - 2\cos x + \frac{1}{2} \left(x + \frac{\sin 2x}{2} \right) \right]_0^{2\pi}$$

$$= \pi \left[2\pi - 2\cos 2\pi + \frac{1}{2} \left(2\pi - \frac{\sin 2 \times 2\pi}{2} \right) - \left(0 - 2\cos 0 + \frac{1}{2} \left(0 - \frac{\sin 0}{2} \right) \right) \right]$$

$$= \pi [2\pi - 2 + \pi + 0 - 0 + 2 + 0]$$

$$= \pi \times 3\pi$$

$$= 3\pi^2 \text{ units}^3$$

1 for correct expression/integrn

1 for correct substitution for $\sin^2 x$

1 for correct integration.

1 for correct answer

b) $\int_0^{\sqrt{2}} \frac{\sqrt{2}}{\sqrt{4-2x^2}} dx = \int_0^{\sqrt{2}} \frac{\sqrt{2}}{\sqrt{2(2-x^2)}} dx$

$$= \int_0^{\sqrt{2}} \frac{1}{\sqrt{2-x^2}} dx$$

$$= \left[\sin^{-1} \frac{x}{\sqrt{2}} \right]_0^{\sqrt{2}}$$

$$= \sin^{-1} \frac{\sqrt{2}}{\sqrt{2}} - \sin^{-1} \frac{0}{\sqrt{2}}$$

$$= \frac{\pi}{2}$$

1 for working

1 for correct integration express.

1 for correct ansn

8

exp. becomes $2 \sin\left(x - \frac{\pi}{6}\right)$

$$\therefore 2 \sin\left(x - \frac{\pi}{6}\right) = 1 \quad 0 \leq x \leq 2\pi$$

$$\sin\left(n - \frac{\pi}{6}\right) = \frac{1}{2}$$

$$\therefore x = \frac{\pi}{6} = \frac{\pi}{6}, \frac{5\pi}{6}.$$

$$\therefore n = \frac{2\pi}{6}, \frac{6\pi}{6}$$

$$= \frac{11}{3}, 11 \quad 0 \leq x \leq 2\pi$$

2 for expression

i. for $x = \frac{\pi}{6} = \frac{\pi}{6}, \frac{5\pi}{6}$

1 for correct answers

8

and $BF = EF$ (from part (i))

so $BF = FD$ (both equal EF)

1 mark for correct conclusion i.e. with reason.

d)

i) Aim: To prove $\widehat{FE\Gamma} = \frac{\pi}{2} - \alpha$

Proof: $\angle AEB = \frac{\pi}{2}$ (angle in a semi-circle)

$\therefore \angle B\hat{K}I = \frac{\pi}{2}$ (supplementary angles / straight line)

and $\hat{BFF} = \angle (BF = EF, \text{ tangents from external point are equal})$

$$\therefore \angle FED = \frac{\pi}{2} - \alpha$$

ii) Now $\angle D = \frac{2\pi}{2} - \frac{\pi}{2} = \alpha$ (angle sum of Δ)

So $F(F) = F(F)$ hence $F(F) = FF$

1 mark for
1 correct statement
with reason.

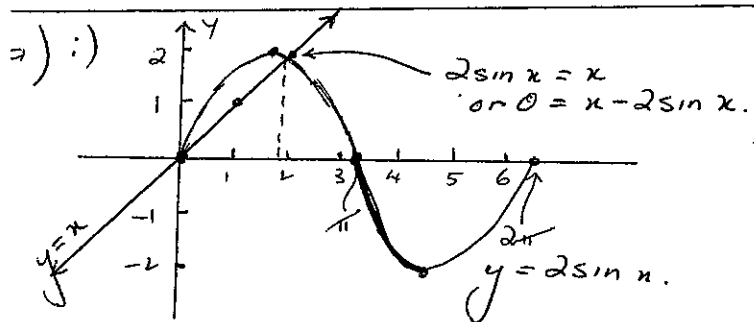
②

for correctly showing result.

1 mark for showing $E \cap F =$

Most students did well in (i) & (ii) only a few did not memorize the formula correctly. (10)

QUESTION 14



ii) $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$

$$x_2 = 2 - \frac{f(2)}{f'(2)}$$

$$= 2 - \frac{(2 - 2 \sin 2)}{(1 - 2 \cos 2)}$$

$$\approx 1.90 \text{ (2 d.p.)}$$

1 for both graphs
1 for suitable size scale, labels, and showing root near $x=2$

1 for correct expression for x_2

1 for correct value.

b) P(2ap, ap²) $x^2 = 4ay$

i) $x^2 = 4ay$

$$y = \frac{x^2}{4a}$$

$$y' = \frac{2x}{4a} = \frac{x}{2a}$$

at $x = 2ap$, $y' = \frac{2ap}{2a} = p$

$\therefore$ normal, $m = -\frac{1}{p}$

equation, $y - ap^2 = -\frac{1}{p}(x - 2ap)$

$$py - ap^3 = -x + 2ap$$

$$x + py = 2ap + ap^3 \quad (Q \in D)$$

1 for derivative, gradient at P

1 for correctly showing equation

QUESTION 14 continued...

b) cont...

ii) meets y axis at Q, $x=0$

$$0 + py = 2ap + ap^3$$

$$y = 2a + ap^2$$

$$\therefore Q = (0, 2a + ap^2)$$

iii) P(2ap, ap²) Q(0, 2a + ap²)

$$R = \left(\frac{2 \times 0 - 1 \times 2ap}{2-1}, \frac{2(2a+ap^2) - ap^2}{2-1} \right)$$

$$= \left(\frac{-2ap}{1}, \frac{4a+ap^2}{1} \right)$$

$$= (-2ap, 4a + ap^2) \quad (Q \in D)$$

iv) $x = -2ap$ $y = 4a + ap^2$

$$p = -\frac{x}{2a}$$

$$\therefore y = 4a + a \left(\frac{-x}{2a} \right)^2$$

$$= 4a + \frac{ax^2}{4a^2}$$

$$y = \frac{x^2}{4a} + 4a, \text{ which is a parabola.}$$

$$\therefore 4ay = x^2 + 16a^2$$

$$x^2 = 4ay - 16a^2$$

$$x^2 = 4a(y - 4a)$$

$$\therefore V = (0, 4a) \quad \text{Focus} = (0, 5a)$$

most students did well in b) i) & (iii)

1 for correct Q coordinates.

1 for correct expression.

1 for correctly showing result.

a few students did not understand that this is the Cartesian equation (in terms of x & y)

1 for locus.

1 for vertex

1 for focus.

QUESTION 14 continued.

$$c) \quad 3x - 5y - 1 = 0, \quad bx - 2y + 2 = 0$$

$$m_1 = \frac{3}{5}$$

$$m_2 = \frac{b}{2}$$

$$\tan 45^\circ = 1$$

$$\therefore \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = 1$$

$$\left| \frac{\frac{3}{5} - \frac{b}{2}}{1 + \frac{3}{5} \cdot \frac{b}{2}} \right| = 1$$

$$\frac{\frac{6-5b}{10}}{\frac{10+3b}{10}} = \pm 1$$

$$\frac{6-5b}{10+3b} = 1 \text{ or } \frac{6-5b}{10+3b} = -1$$

$$6-5b = 10+3b \text{ or } 6-5b = -(10+3b)$$

$$-4 = 8b$$

$$b = -\frac{1}{2}$$

$$16 = 2b$$

$$b = 8.$$

Most students did well, a few did not memorize formula correctly

1 for correct expression

1 for $b = -\frac{1}{2}$

1 for $b = 8.$